

Aidan Adams



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Education

Virginia Polytechnic Institute and State University

M.Eng. in Computer Science, AI/ML Focus (GPA: 3.84 / 4.00)

May 2026

Alexandria, VA

Virginia Polytechnic Institute and State University

B.S. in Computer Science (GPA: 3.68 / 4.00)

May 2025

Blacksburg, VA

Experience

Peraton Labs

Senior Associate Research Scientist

Jun 2026 – Present

Remote

Peraton Labs

AI Research Science Intern

Aug 2025 – May 2026

Herndon, VA

- Implemented a social media agent allowing users to query millions of social media documents through natural language via agentic control of the Babel Street API. Optimized search results with similarity metrics.
- Developed resilient skills-based agentic workflows with Temporal.io for reproducible custom tooling.
- Architected and deployed an S3-based data lake as a scalable plugin to support terabytes of user documents.

Peraton

Software Engineering Intern

Jan 2025 – Aug 2025

Herndon, VA

- Built Python based LLM agents with Retrieval Augmented Generation (RAG), tool calling, and iterative workflows.
- Engineered a persistent memory system using PostgreSQL to enable agents to recall user-specific data and tailor responses with long term context. Utilized diff-based rewrites, reducing token count by up to 90%.
- Developed analytics infrastructure to track token usage, cost, and user behavior across the LLM platform, supporting customer budgeting and system monitoring. Enabled usage-based account throttling to prevent system abuse.
- Created a corresponding analytics agent that converts user prompts into SQL queries, outputs to CSV, and executes python to create insights and generate graph visualizations, saving customers thousands of dollars in tokens.

BrightVine Solutions

Software Engineering Intern

May 2023 – Aug 2023, May 2024 – Aug 2024

Remote

- Built an Azure OpenAI integrated chatbot that converts natural language to SQL with client specific fine-tuning.
- Designed and deployed a batch import error handling system using SQL, C#, and XML to production.

Virginia Tech Department of Computer Science

Teaching Assistant, CS 1064: Intro to Python

Jan 2023 – Dec 2024

Blacksburg, VA

- Supported 2000+ students by mentoring, debugging, and answering questions on foundational Python coding.

Projects

Smart Home os | FastAPI, Redis, MQTT, PostgreSQL, React, WebSockets, Docker

- Developed a digital twin backend with Redis to cache smart device states, achieving sub-millisecond API read times.
- Built a custom AI orchestrator with persistent conversational memory and parallel tool calling suites to securely control physical IoT gateways and read local state in real time, with tools, and context tailored to each user account.
- Containerized full stack with Docker Compose, integrating a React dashboard updated instantly via WebSockets.

AI Deepfake Detection System | Neural Networks, Computer Vision, SVM, AWS, API, Next.js, Flask

- Team Lead on AI deepfake detection capstone project with 86% accuracy utilizing SVM's and CNN's.
- Managed large data size with cloud resources for training while iteratively testing different models effectiveness.

Q-Learning Tag Agents | Q-Learning, OpenAI Gymnasium, Matplotlib, PyGame

- Built a custom multi-agent Tag environment in Gymnasium with Q-learning, featuring discrete 8-direction movement and dynamic epsilon-greedy exploration, visualized in PyGame.

Technical Skills

Languages: Python, C, Java, C#, SQL, JavaScript, TypeScript **Tools:** FastAPI, Flask, React, Next.js, Redis, TensorFlow, PyTorch, Pandas, Scikit-Learn **Cloud/DevOps:** Docker, Azure Blob Storage, AWS S3, CI/CD, Linux, API Deployment **Concepts:** LLM Agents, RAG, NLP, Data Engineering, API Design, WebSockets, Systems Design, Agile